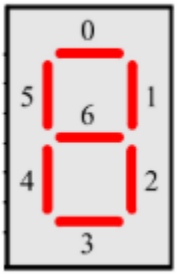
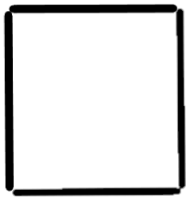
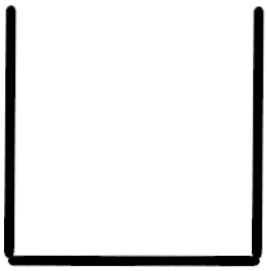
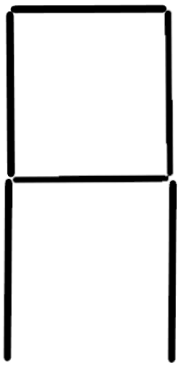
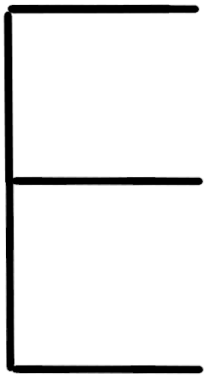
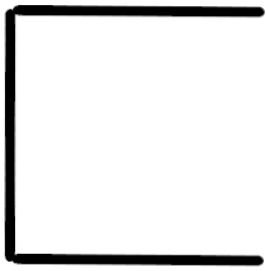
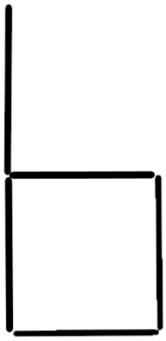
Dylan Thornburg
9/25/2025

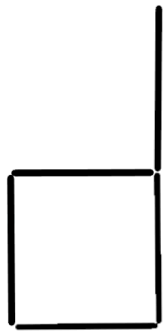
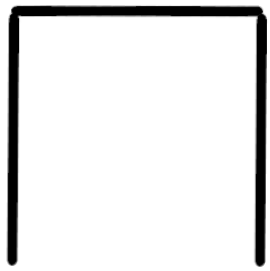
CSEN 127 Prelab1

1.



num	3-bit	7-bit	Symbol
0	000	1011100	o (2,3,4,6) 
1	001	0011100	u (2,3,4) 
2	010	1110111	A (0,1,2,4,5,6) 

3	011	1111001	E (0,3,4,5,6) 
4	100	1011000	c (3,4,6) 
5	101	1111100	b (2,3,4,5,6) 
6	110	1011110	d (1,2,3,4,6)

			
7	111	1010100	n (2,4,6) 

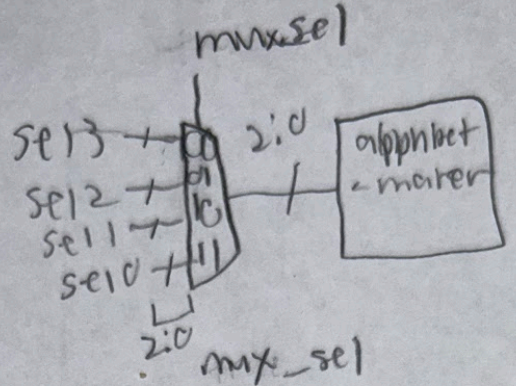
2.

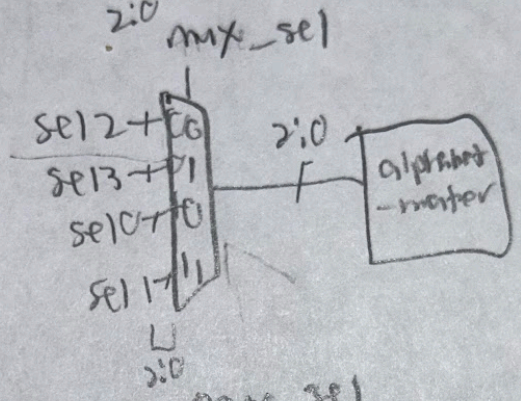
```
module alphabet_maker (
        input  wire [2:0] sel,
        output reg  [6:0] segment
);
        always @* begin
                case (sel)
                3'b000: segment = 7'b1011100;
                3'b001: segment = 7'b0011100;
                3'b010: segment = 7'b1110111;
                3'b011: segment = 7'b1111001;
                3'b100: segment = 7'b1011000;
                3'b101: segment = 7'b1111100;
                3'b110: segment = 7'b1011110;
                3'b111: segment = 7'b1010100;
                default: seg = 7'b0000000;
                endcase
        end
endmodule
```

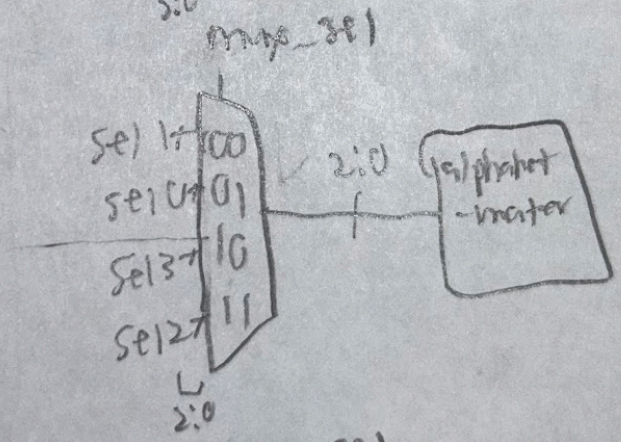
3.

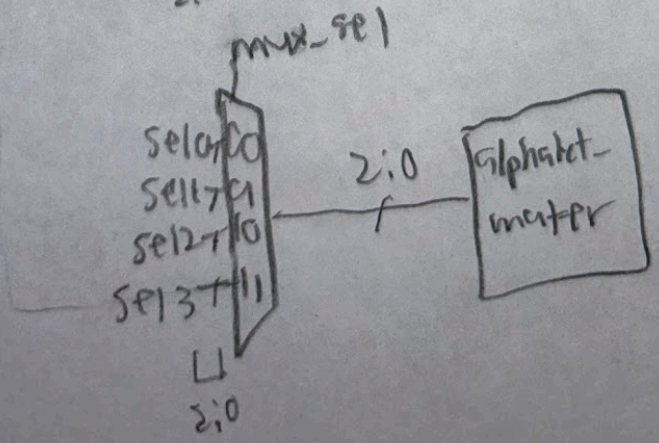
```
module 4_1_mux(
	input [2:0] sel0,
	input [2:0] sel1,
	input [2:0] sel2,
	input [2:0] sel3,
	input [1:0] mux_sel,
	output [2:0] out
);
	assign out = mux_sel[1] ? (mux_sel[0] ? sel3 : sel2) : (mux_sel[0] ? sel1 : sel0);
endmodule
```

4.


mux_sel
sel3
sel2
sel1
sel0
00
01
10
11
2:0
2:0
alphabet
-meter


mux_sel
sel2
sel3
sel0
sel1
00
01
10
11
2:0
2:0
alphabet
-meter


mux_sel
sel1
sel0
sel3
sel2
00
01
10
11
2:0
2:0
alphabet
-meter


mux_sel
sel0
sel1
sel2
sel3
00
01
10
11
2:0
2:0
alphabet-
meter